Spectral Reconstruction From Satellite Multispectral Imagery Using Convolution and Transformer Joint Network

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***Abstract*— Spectral reconstruction (SR) based on satellite multispectral (MS) images can produce high spatial resolution hyperspectral (HS) images at a reasonable cost, significantly expanding the application of satellite-based HS remote sensing. As a challenging ill-posed problem, existing methods have difficulty making full use of local and global information of space and spectra to guide the reconstruction, resulting in limited accuracy in large-scale scenes with complex ground features and severe spectral mixing. In this article, we propose a novel convolution and transformer joint network (CTJN) to address the challenge of high-accuracy SR in complex scenes. The CTJN is cascaded with shallow feature extraction modules (SFEMs) and deep feature extraction modules (DFEMs), which can explore local spatial features and global spectral features. Besides, a high-frequency transformer block (HF-TB) is designed to highlight the detailed features of the images to prevent significant high-frequency information loss, which could improve the reconstruction results in regions with drastic feature changes. Moreover, a spatial–spectral recalibration block (SSRB) is proposed to perform explicit constraints on the reconstructed points by exploiting the correlation among neighboring pixels and adjacent spectra. Extensive experimental results on four HS–MS datasets and one MS dataset demonstrate that the proposed CTJN outperforms the state-of-the-art methods in large- and small-scale scenes.**

***Index Terms*— Convolution neural network, hyperspectral (HS) imagery, multispectral (MS) imagery, spectral reconstruction (SR), spectral super-resolution, transformer.**

I. Introduction

HYPERSPECTRAL (HS) satellites are very effective tools for remote sensing and Earth observation. Characterized by their rich and discriminatively spectral information, HS images can accurately distinguish different land cover classes and are widely used in tasks, such as object detection [1], [2], [3], anomaly detection [4], [5], [6], fine classification [7], [8], [9], and scene segmentation [10], [11]. However, due to the limitations of sensor techniques, HS satellite images have poor spatial resolution and severe spectral mixing. In addition, the long revisiting period and the small number of HS satellites make it challenging to efficiently obtain high spatial resolution HS images. These issues severely restrict the application of the HS data. Comparatively, multispectral (MS) satellites have a higher spatial resolution and a shorter revisiting period, which help overcome the drawbacks of HS satellite applications [12].

Several approaches have been developed to perform spatial super-resolution to obtain satellite HS images with high spatial resolution [13], [14], [15]. Besides, other studies tried to implement spectral super-resolution by fusing HS images with MS images [16], [17], [18]. All these methods can effectively improve the spatial resolution of HS images. However, none of the methods mentioned, which all require raw HS images as input, overcome the issue that satellite HS images are costly to acquire. Li et al. [12] and Li and Gu [19] led the research of satellite MS to HS reconstruction and proposed the progressive spatial–spectral joint network (PSJN) and multitemporal deep learning network for the reconstruction of single and multitemporal HS images, respectively. As a result, HS images with high spatial resolution and sizable views can be obtained based on actual satellite MS data.

Spectral reconstruction (SR) is a seriously ill-posed problem that aims to recover hundreds of HS bands from a few MS bands. Previous research relied on matrix mapping, which is rapid for reconstruction but inaccurate [20], [21], [22]. Subsequently, the sparse representation-based approaches utilize prior knowledge to improve SR accuracy greatly [23], [24], [25], [26]. However, these methods require complex computations and impose restrictions on the scene's content, making them unsuitable for large and complex scenes. Due to the powerful feature extraction capability of deep learning, some researchers have recently applied it to SR and achieved improved results [19], [27], [28], [29], [30], [31], [32], [33], [34], and it is gradually becoming the main method for SR.

In the computer vision field, there have been various studies on spectral super-resolution of input RGB images based on deep learning, and the processing methods can be roughly divided into two classes based on convolution neural network (CNN) [35], [36], [37], [38] and generative adversarial network (GAN) [39], [40], [41], [42]. Recently, some transformer-based SR methods have been proposed and achieved better results [43], [44].

It is still challenging to perform accurate SR in large-scale scenes with complex ground features. First, the limited spatial resolution, wide spatial observation width, and severe spectral mixing hinder the performance of SR. Besides, the reconstructed spectral curves by existing methods are distorted

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in regions with drastic feature changes. Analysis shows that most existing methods still make it difficult to fully use local and global information of space and spectra to guide the reconstruction [45], [46] and do not focus on the rich high-frequency information in complex scenes, resulting in limited accuracy in large complex scenes.

To address the aforementioned issues, a novel convolution and transformer joint network (CTJN), which combines 3-D-CNN and transformer to correlate local spatial information with global spectral information, is proposed for high-accuracy SR in large-scale scenes with complex ground features. First, it is noted that the CNN-based methods have difficulty capturing the global correlation of spectra [46], which could lead to errors accumulating in complex scenes, and we use a self-attention mechanism in the spectral dimension to capture the interconnections between remote bands through attention maps while simultaneously collecting spatial and spectral fine details using 3-D-CNN. Second, a high-frequency transformer block (HF-TB) is implemented to prevent significant loss of detailed information, which could improve the reconstruction results in regions with drastic feature changes. Finally, a spatial–spectral recalibration block (SSRB) is built with prior knowledge to reweight the feature maps in both spatial and spectral dimensions. Comprehensive experimental results demonstrate that our proposed network achieves optimal performance. The main contributions of this article could be summarized as follows.

1) A CTJN is proposed for SR in large-scale scenes with complex ground features, which can extract local spatial features and global spectral features. To the best of our knowledge, this is the first effort to integrate 3-D-CNN with transformer for SR in the remote sensing field.

2) An HF-TB is introduced to alleviate the problem of severe distortion of spectral curves in regions with drastic feature changes, which could highlight the high-frequency information and keep the images from being overly smooth.

3) An SSRB is designed by combining prior knowledge with data-driven methods, which perform explicit constraints on the reconstructed points by exploiting the correlation among neighboring pixels and adjacent spectra.

The remainder of this article is structured as follows. Section II briefly describes the evolution of SR methods. In Section III, we present the framework of the proposed CTJN. The details and analysis of the extensive experiments are presented in Section IV. Finally, we make the conclusions in Section V.

II. Related Work

SR methods can be roughly categorized into knowledge-driven methods and data-driven methods. Most knowledge-driven methods build the relationship between the input MS images and the output HS images utilizing prior information [24], [26], [47]. The majority of data-driven methods utilize enormous amounts of data to discover the latent relationship between the inputs and the outputs for accurate SRs [19], [27], [29].

The methods based on knowledge-driven can be divided into two types: matrix mapping-based methods and sparse representation-based methods. The matrix mapping-based methods received attention from early researchers due to lower computing costs. The representative approaches are principal component analysis (PCA) and pseudoinverse (PI) [21], [22]. As the hardware advances, these methods are gradually supplanted. The sparse representation-based methods are currently the predominant methods based on knowledge-driven. Arad and Ben-Shahar [48] applied sparse encoding to acquire the spectral dictionary, but this approach failed to account for spatial restrictions, resulting in low reconstruction quality. Fotiadou et al. [23] proposed a coupled dictionary learning model based on joint feature space to improve the reconstructed performance. Yi et al. [24] improved the reconstruction results by constraining the spatial and spectral components separately. Gao et al. [26] achieved SR by learning low-rank dictionaries from the overlapping regions of HS and MS images. Hong et al. [47] reconstructed high-quality HS images by utilizing locally constrained relationships. However, these methods impose more restrictions on the scenes [49], [50], which are not always met in large-scale scenes with complex features. Furthermore, these methods do not delve deeply into the interconnections among data, resulting in the reconstructed HS images with poor accuracy in complex scenes.

Deep learning-based methods for SR in remote sensing are gradually being presented due to their superior performance [51], [52]. It can broadly be split into two types according to the reconstruction process. The first is considering each pixel as a combination of multiple basis functions. Liu et al. [33] and [34] performed SR based on spectral unmixing and a spectral library while making further assumptions about the scenes, e.g., there are only a few different types of materials in scenes and the same materials have the same reflectance. However, given the wide spatial coverage and limited spatial resolution of satellite remote sensing, the restrictions may not be reasonable. The second is directly learning the mapping relationship between MS and HS. Mei et al. [28] proposed a spatial and spectral joint super-resolution network (SSJSRN) by 3-D convolution to extract spatial and spectral local features. Zheng et al. [27] introduced a spatial–spectral residual attention network (SSRAN) to learn spatial and spectral neighboring relations by 2-D convolution and 1-D convolution. Han et al. [31] combined clustering with neural networks to generate HS images. Mu et al. [29] attempted SR between two MS sensors using transformer by learning global spatial information. Liu et al. [32] used GANs to perform SR without introducing spectral response prior. Li and Gu [19] suggested a PSJN by utilizing the progressive reconstruction strategy and achieved good results in large-scale scenes. Most methods have achieved leading results on some datasets, such as Indian Pines (IP), Pavia University (PU), and Washington DC Mall (WDCM). However, these methods perform the reconstruction with local information from space and spectra, which could lead to the accumulation of reconstruction errors in large-scale scenes with complex ground features.

Spectral super-resolution in computer vision is similar to SR in remote sensing. The task is to input RGB images and output

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HS images of 31 bands from 400 to 700 nm. Since 2017, methods based on deep learning for spectral super-resolution have gradually been proposed [34], [35], [43], [44], [53], [54], [55], [56], [57]. Also, benefiting from the new trends in image restoration and enhancement (NTIRE) SR challenge, some excellent deep learning methods showed fantastic potential. Shi et al. [53] designed an HSCNN by using a density connection structure and won the 2018 challenge. Li et al. [54] won the "clean" track in the 2020 challenge by proposing an adaptive weighted attention network (AWAN) with the camera spectral sensitivity loss function. Cai et al. [43] applied the transformer structure to SR and won the 2022 challenge. Gao et al. [26] and Zheng et al. [27] demonstrated that spectral super-resolution in computer vision is distinct from SR in remote sensing. Therefore, directly applying RGB spectral super-resolution to the remote sensing field may result in suboptimal performance.

III. Methodology

A. Problem Formulation

The energy value recorded by the sensor is affected by solar radiation, atmospheric absorption, sensor performance, and other factors; it could be denoted as

$$\mathrm{DN}_i(p) = \int K_i(\lambda) I(\lambda) R(p,\lambda) S_i(\lambda) d\lambda + N_i \quad (1)$$

where $\mathrm{DN}_i(p)$ is the energy value of the pth region recorded by the ith sensor. $K_i(\lambda)$ represents the influence of the sensor, including the spectral transmittance of the lens and the detection efficiency of the sensor. $I(\lambda)$ is the value of solar radiation. $R(p,\lambda)$ is the reflectance of the pth region. $S_i(\lambda)$ denotes the ith sensor's spectral response function (SRF), and $\int S_i(\lambda)d\lambda = 1$, N_i is noise.

After radiometric calibration, it can be considered that the influence of sensor differences is eliminated, Thus, (1) could be equivalent to

$$\mathrm{DN}_i(p) = \int L(p,\lambda) S_i(\lambda) d\lambda \quad (2)$$

where $L(p,\lambda)$ is the radiation energy at λ wavelength on the pupil of the lens, which could be formulated as

$$L(p,\lambda) = I(\lambda) R(p,\lambda). \quad (3)$$

The HS images are obtained by scanning the ground with a series of sensors with very narrow bandwidths. The recorded energy value can be considered the energy value at the central wavelength of the sensor, which can be expressed as

$$\begin{aligned} H_i(p) &= \int L(p,\lambda) S_i(\lambda) d\lambda \\ &\cong L(p,\lambda_i) \int S_i(\lambda) d\lambda = L(p,\lambda_i) \end{aligned} \quad (4)$$

where λ_i is the central wavelength of the ith HS sensor. Therefore, it can be considered that the spectral curves recorded by the HS sensors are discretization of the continuous spectral curves.

The MS images are obtained by scanning the ground with several wide bandwidth sensors, and the energy value recorded by each sensor is correlated with the radiation energy and SRF. It could be expressed as

$$M_k(p) = \int L(p,\lambda) S_k(\lambda) d\lambda \cong \sum_{i=1}^{N} H_i(p) S_k(\lambda). \quad (5)$$

Equation (5) is equivalent in a matrix form as

$$\mathbf{M} = \mathbf{SH} \quad (6)$$

where $\mathbf{M} \in \mathbb{R}^{K\times 1}$ is the value of MS, $\mathbf{H} \in \mathbb{R}^{N\times 1}$ is the value of HS, and $\mathbf{S} \in \mathbb{R}^{K\times N}$ is the spectral response matrix. For the inverse transformation of (6), it could be expressed as

$$\mathbf{H} = f(\mathbf{S}, \mathbf{M}; \theta). \quad (7)$$

When $K \ll N$, (7) is a seriously underconstrained ill-posed problem. Arad and Ben-Shahar [48] demonstrated that spectral recovery is feasible when the signal is confined to a low-dimensional manifold in high-dimensional space and the frequency of the metamers in the manifold is low. We transform (7) into a supervised optimization problem, which could be expressed as

$$\arg\min_\theta \mathcal{L}(f(\mathbf{S}, \mathbf{M}; \theta), \mathbf{H}). \quad (8)$$

In large scenes with complex ground features, considering only neighborhood information will lead to error accumulation. Therefore, we consider both local spatial information and global spectral information in designing a CTJN model for large-scale scenes with complex features. Simultaneously using the errors between MS images and HS images as loss functions, this could be expressed as

$$\arg\min_\theta (\mathcal{L}_{\mathrm{MRAE}}(f(\mathbf{S}, \mathbf{M}; \theta), \mathbf{H}) + \mathcal{L}_{\mathrm{MRAE}}(\mathbf{S} f(\mathbf{S}, \mathbf{M}; \theta), \mathbf{M})). \quad (9)$$

B. Network Architecture

The overall structure of the network is shown in Fig. 1. The proposed CTJN consists of two main parts, the shallow feature extraction modules (SFEMs) and deep feature extraction modules (DFEMs). First, for the input MS images $I_{\mathrm{MSI}} \in \mathbb{R}^{c\times H\times W}$, $F \in \mathbb{R}^{C\times H\times W}$ are obtained by a 1×1 convolution, where $H \times W$ denotes the spatial size of the images, c is the number of MS bands, and C is the number of HS bands. Next, the N_F SFEMs are applied to extract spatial and spectral shallow features $F_{\mathrm{SFEM}} \in \mathbb{R}^{C\times H\times W}$; each SFEM consists of a spatial–spectral conv3-D module (SCM) with a residual connection, and it could be expressed as

$$F_{\mathrm{SFEM}} = H_{\mathrm{SCM}}(F) + F \quad (10)$$

where F is the input of the SFEM and $H_{\mathrm{SCM}}(\cdot)$ represents the system response of SCM. Subsequently, the N_D DFEMs are proposed to extract spatial and spectral deep features $F_{\mathrm{DFEM}} \in \mathbb{R}^{C\times H\times W}$; each DFEM consists of a U-shaped spectral transformer module (USTM) with a residual connection, and it could be calculated as

$$F_{\mathrm{DFEM}} = H_{\mathrm{USTM}}(F) + F \quad (11)$$

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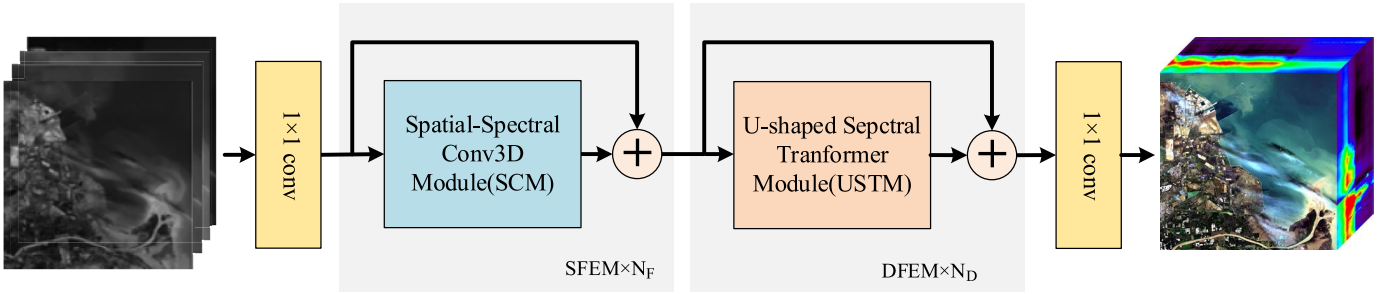


Fig. 1. Overall structure of CTJN for SR.

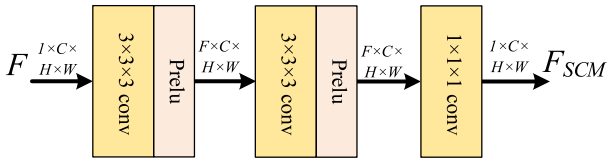


Fig. 2. Structure of SCM.

where F represents the input of the DFEM and $H_{\mathrm{USTM}}(\cdot)$ stands for the output of USTM. Finally, a 1×1 convolution is applied to reconstruct the features and output the HS images.

By cascading SFEMs and DFEMs, the network can first learn local spatial and spectral information and then learn global spectral information. It could prevent the model from directly capturing intercorrelations between long-range bands and ignoring detailed information, keeping the images from being overly smooth.

C. Spatial–Spectral Conv3-D Module

Fig. 2 shows that the proposed SCM is comprised of multiple 3-D convolutional layers. 3-D convolution kernels can capture both spatial and spectral features without introducing an excessive number of parameters. To avoid oversmoothing the images, we select tiny convolution kernels to extract finer spatial and spectral features.

The 3-D convolution kernels can be defined as $W^i \in \mathbb{R}^{N\times C\times H\times W}$, where N is the number of 3-D kernels and C, H, and W represent the size of the kernel in 3-D. For the given feature maps $F \in \mathbb{R}^{C\times H\times W}$, an additional feature dimension is needed to match the 3-D convolution, and the maps can be expressed as $F \in \mathbb{R}^{F\times C\times H\times W}$ (F is 1 in the input phase). The process of 3-D convolution can be expressed as

$$\begin{aligned} F^i_{\mathrm{output}} &= f\left(W^i \times F^i_{\mathrm{input}} + \delta^i\right) \\ F^{i+1}_{\mathrm{input}} &= F^i_{\mathrm{output}} \end{aligned} \tag{12}$$

where the subscript "i" represents the operation in the ith convolutional layer, $f(\cdot)$ is the activation function, and δ^i is the bias. We use two cascaded $3 \times 3 \times 3$ convolutions to concurrently capture spatial and spectral fine texture details, followed by a $1 \times 1 \times 1$ convolution that is applied to remove the redundant F dimension to increase the network's scalability.

D. U-Shaped Spectral Transformer Module

The self-attention mechanism allows the model to obtain global information about the spectrum by getting the correlation among long-range channels with a shallow network, which is difficult to achieve with CNN alone. We combine the self-attention mechanism with U-Net to form USTM, which could use global information from the spectrum to guide the reconstruction and reduce the accumulation of errors.

The overview pipeline of USTM is shown in Fig. 3(a). The transformer blocks are used to extract effective features from the top to the bottom of the U-shaped structure. In the encoder stage, multiple HF-TBs are used to extract abstract features at each level while progressively performing downsampling operations to decrease the spatial resolution and increase the number of channels. We use multiple raw transformer blocks (RTBs) to further extract abstract features during the bottleneck stage. In the decoder stage, a fully symmetric structure is adopted with the encoder, and the RTBs are applied to integrate features at each level. Besides, the features of the encoder are concatenated to the decoder via skip connections after SSRB to avoid effective information loss during the downsampling process. We employ the pixel-unshuffle and pixel-shuffle operations [58] to implement downsampling and upsampling. Finally, a 3×3 convolution is applied to fine-tune the feature maps.

E. High-Frequency Transformer Block

The encoder maps the input features to a low-dimensional latent space, and the decoder recovers them as high-dimensional features. In this procedure, high-frequency information is inevitably lost due to the downsampling operation, resulting in reconstructed images that tend to be smooth and spectral curves severely distorted in regions with drastic ground feature changes.

To address this issue, as shown in Fig. 3(b), an HF-TB is proposed in the encoder stage. Compared with the RTB [as shown in Fig. 3(c)], a High_fre block is inserted between the spectral position-sensitive self-attention (SPS-SA) and the feed forward network (FFN) to emphasize the detailed information of the feature maps. It is a learnable block, and the specific structure is shown in Fig. 4. For the given feature maps F_{in}, our High_fre block first generates the downsampled maps by the average-pooling operation. Next, we resize the feature maps to the same size as F_{in} using bilinear interpolation. After differencing, we obtain the residual feature maps rich

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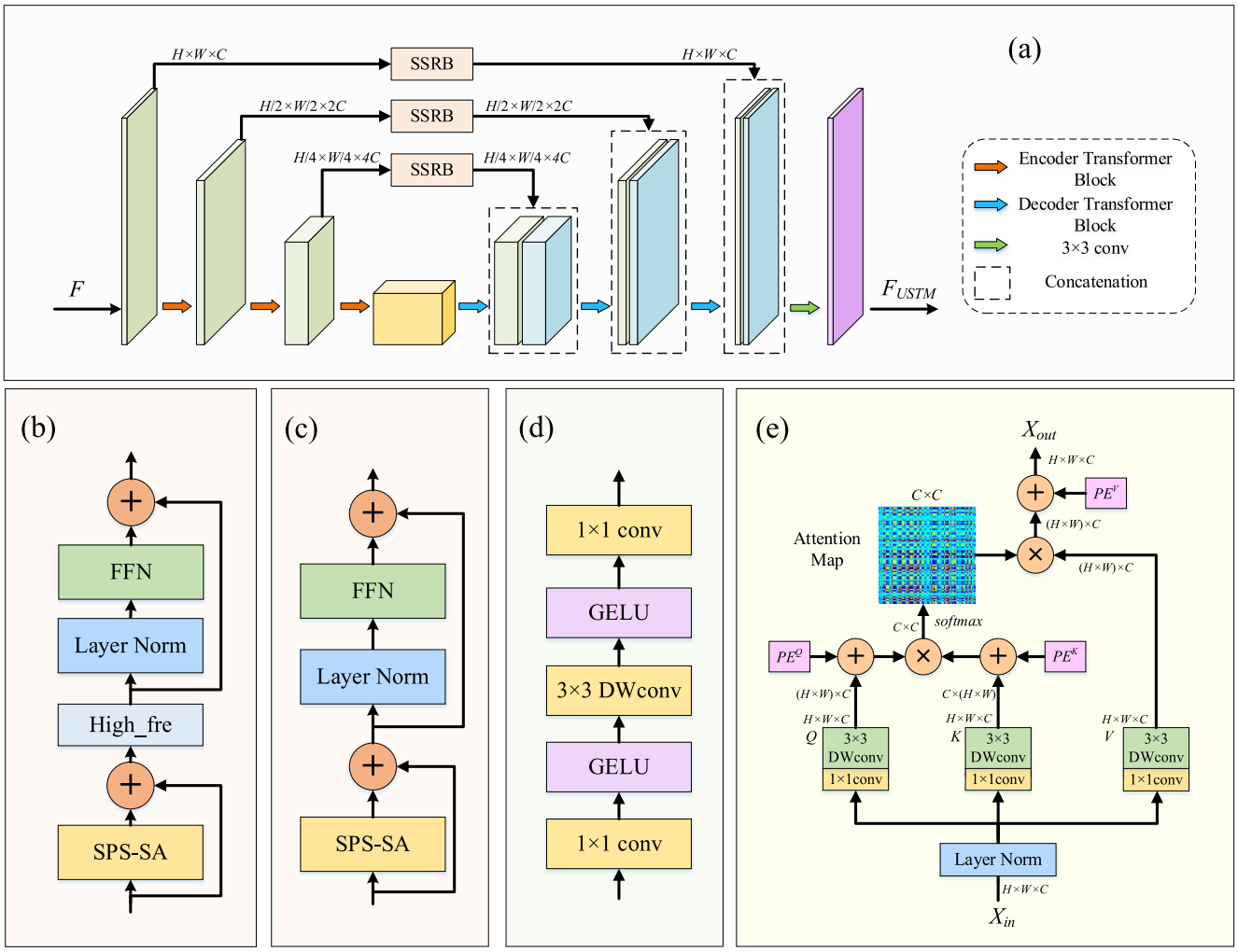


Fig. 3. Framework of the proposed USTM. (a) Overview of USTM. (b) Structure of HF-TB. (c) Structure of RTB. (d) Structure of FFN. (e) Structure of SPS-SA.

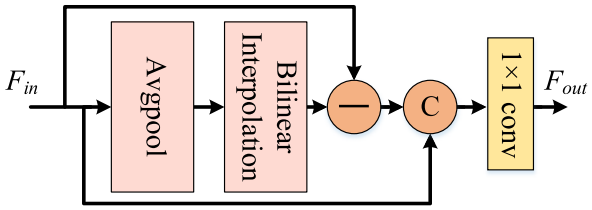


Fig. 4. Structure of High_fre block.

in high-frequency information. Finally, the feature maps F_{out} are obtained with a 1×1 convolution to adjust the dimension. Therefore, the high-frequency information can be better focused in the encoder, and the information can be delivered to the decoder via skip connections, which is beneficial for preserving the detailed features and has good applicability for large complex scenes.

F. Spectral Position-Sensitive Self-Attention

The transformer has been applied to SR previously [43], [59]. Compared to traditional transformers such as ViT [60], these approaches can create long-range dependencies in spectral dimension while reducing computational complexity (linear complexity). Inspired by these methods [43], [59], [61], we design the SPS-SA by embedding the position information of Q, K, and V, as shown in Fig. 3(e). It could be formulated as

$$\begin{aligned} \text{Attn} &= \text{softmax}\big((K + \text{PE}(K))^{\text{T}}(Q + \text{PE}(Q))\big) \\ X_{\text{out}} &= W \times V \times \text{Attn} + \text{PE}(V) \end{aligned} \tag{13}$$

where $Q, K, V \in \mathbb{R}^{H \times W \times C}$ are the linearly projected values of the input X_{in}, yielding $Q = W^Q X_{\text{in}}$, $K = W^K X_{\text{in}}$, and $V = W^V X_{\text{in}}$ which represent query, key, and value projections, respectively. PE$(\cdot)$ represents the position embedding function. It is considered that the generated K, Q, and V all contain position information. Therefore, in contrast to networks such as ViT [60], we use K, Q, and V as input, and after PE$(\cdot)$, which is composed of two 3×3 convolutional layers, the position embedding could be output. The added information about the position of K and Q allows the generated attention maps to notice the relative position relationship within the spectrum, which can better integrate spectral-wise interdependencies and help the network complete accurate reconstruction. In practice,

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we use multihead self-attention to generate different attention maps simultaneously.

G. Spatial–Spectral Recalibration Block

It is crucial to improve the correlation of neighboring space and adjacent spectra of the feature maps with prior knowledge. This can help the network converge quickly and improve the accuracy of the reconstructed HS images [62], [63]. The SSRB is proposed to use prior knowledge to enhance the reconstruction capability of the network. The structure of SSRB is shown in Fig. 5. For the given feature maps X_{in}, F_{in} is obtained by a 3×3 convolution. The 3×3 convolution is applied to further extract the semantic information of the features and increase the module's portability. Subsequently, the dual-branch structure is applied to recalibrate the feature maps' weights along the spectral and spatial directions.

In the spectral direction, SSRB first generates two features with $1 \times 1 \times C$ dimension by a global max-pooling operation and a global average-pooling operation. Then, we fuse the features followed by a 1-D convolution to adjust the aggregate information based on the high correlation of adjacent spectra. After the multilayer perceptron (MLP) function adaptively calibrates the response of the features, the spectral features are output after the sigmoid function. It can be expressed as

$$F_{\text{spe}} = \delta(\text{MLP}(W_{1D}(Y_{\text{GAP}}(F_{\text{in}}) + Y_{\text{GMP}}(F_{\text{in}})))) \tag{14}$$

where Y_{GAP} represents the global average-pooling operation, Y_{GMP} stands for the global max-pooling operation, and $\text{MLP}(\cdot)$ represents the multilayer perceptron function, which consists of two 1×1 convolutional layers.

In the spatial direction, SSRB generates two $H \times W \times 1$ dimensional features by a channel max-pooling operation and a channel average-pooling operation. After the feature fusion, we use a 3×3 convolution to extract the local spatial features, which can integrate effective information via the spatial neighboring pixel similarity. Also, the spatial features are adopted after the sigmoid function. It can be formulated as

$$F_{\text{spa}} = \delta(W_{3\times3}(\text{Concat}(Y_{\text{CAP}}(F_{\text{in}}), Y_{\text{CMP}}(F_{\text{in}})))) \tag{15}$$

where Y_{CAP} denotes the channel average-pooling operation and Y_{CMP} represents the channel max-pooling operation. Finally, the outputs X_{out} are acquired following the operation as follows:

$$X_{\text{out}} = X_{\text{in}} + F_{\text{in}} \cdot \left(F_{\text{spe}} \cdot F_{\text{spa}}\right) \tag{16}$$

where "$\cdot$" represents the element-wise product.

In the spectral dimension, we use the addition operation to fuse the features due to the design of 1-D convolution to explicitly aggregate the information of the adjacent spectra. The concatenation operation could reduce the high correlation of the adjacent spectra, which will negatively impact the performance of the subsequent 1-D convolution. In the spatial dimension, we use the concatenation operation to fuse the features because the concatenation operation does not affect the aggregation of the 3×3 convolution for neighboring space and can preserve the different information extracted by max pooling and average pooling more completely, making the network perform better.

The SSRB is proposed to use 1-D convolution in the spectral branch to emphasize the influence of the adjacent spectra and 3×3 convolution in the spatial branch to emphasize the influence of the neighboring space. Compared with squeeze and excitation (SE) [64], efficient channel attention (ECA) [65], and convolutional block attention module (CBAM) [66], the proposed SSRB has a better performance and interpretability. The specific differences between these modules will be demonstrated in the ablation analysis.

TABLE I
DETAILS OF DATASETS

Dataset sign	Satellite	Image size	Spatial resolution
HS200501	ZY102D	1720 × 1634 × 77	30 m
HS200628	ZY102D	1678 × 1656 × 77	30 m
HS200724	ZY102D	1956 × 1588 × 77	30 m
HS200926	ZY102D	1490 × 1469 × 77	30 m
HS210605	ZY102D	1960 × 1584 × 77	30 m
MS200628	GF1WFV	1299 × 1530 × 4	16 m

IV. EXPERIMENTS

A. Datasets

1) Yellow River Delta Reserve and Nearby Towns: Five HS images and one MS image were acquired over the Yellow River Delta Reserve and nearby towns. The HS images are from the ZY102D sensor with a 30-m spatial resolution and 166 spectral bands. A total of 77 bands from 400 to 1050 nm are selected. The MS image is from the GF1 sensor with a wide field of view (WFV) with a 16-m spatial resolution and four spectral bands. For quantitative assessment, as explored in [19], [27], and [28], the ideal MS data utilized in the training and test phases are obtained from the matching HS data using SRF. The corresponding SRF of ZY102D sensor and GF1 WFV sensor is shown in Fig. 6 (after normalization).

The pseudo-color images after atmospheric correction and orthorectification are shown in Fig. 7. These images were collected on different dates, and the details are shown in Table I. To facilitate expression, we use HS200501 to represent an HS image acquired by the ZY102D sensor on May 1, 2020, and use MS200628 to stand for an MS image acquired by the GF1 WFV sensor on June 28, 2020, and so on. Each pixel contains dozens of endmembers because the spatial resolution of HS images is 30 m. It is a challenging task for SR in large-scale scenes with complex ground features and severe spectral mixing.

Three experiments are conducted to extensively evaluate the efficiency of different methods based on similarity. The first experiment (experiment 1) is based on HS200628, which is an image of the Yellow River Delta Reserve. The main scenes in the image are the sea, bare soils, and plants. As shown in Fig. 7(b), the subregions with a size of $1678 \times 829 \times 77$ are chosen for training (red regions), and the remaining regions are used for test. The second experiment (experiment 2) is based on HS210605, which is an image of the Yellow River Estuary

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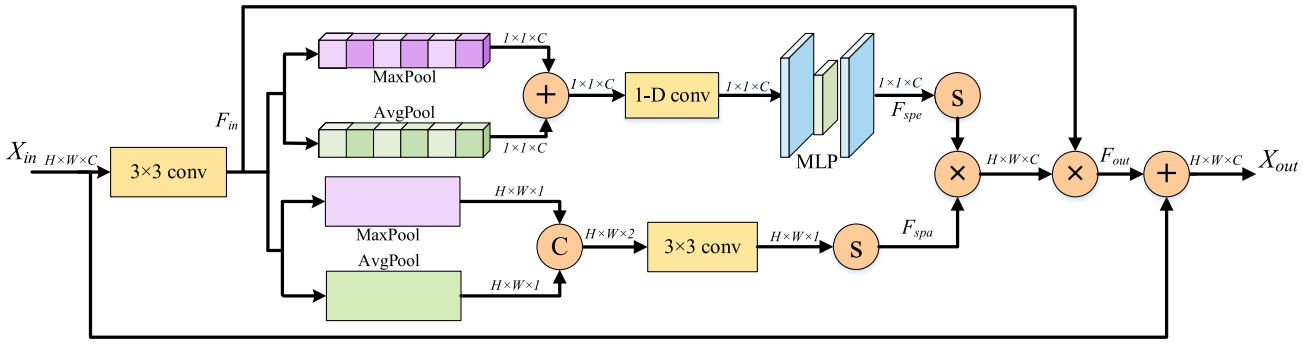


Fig. 5. Structure of SSRB.

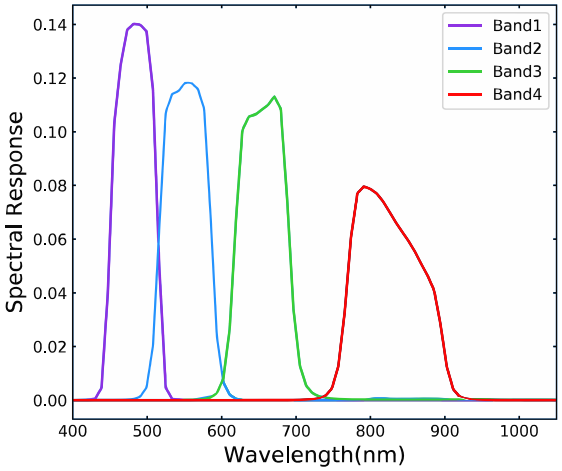


Fig. 6. Normalized SRF from ZY102D to GF1.

TABLE II
GROUND TRUTH ON THE MS200628 DATASET

	Water	Reed	Mudflat	Tamarix	Suaeda	Spartina	Total
Train	82	94	85	88	88	84	521
Test	801	966	903	917	886	866	5339

Town. The main scenes in the image are bare soils, plants, and rivers. The subregions of 1960 × 400 × 77 are picked for training (red regions), and the remaining regions are used for test, as shown in Fig. 7(e). The third experiment (experiment 3) is based on five HS images. Four images acquired in 2020 are used for training, and HS210605 is applied for test. During the process, we mask the thick clouds and perform no processing of the thin clouds. In addition, on the MS200628 dataset, we evaluate the performance of different methods trained in the three experiments based on classification. The ground truth collected for training and testing in the classification experiment is shown in Fig. 8(a) and (b). It contains a total of six classes, including water, reed, mudflat, tamarix, suaeda, and spartina. Table II lists the number of different samples used for training and testing.

2) Pavia University (PU): The PU data were acquired at PU, Italy. The HS image is from the ROSIS sensor with a 1.3-m spatial resolution and 103 bands from 430 to 860 nm. It is a small-scale scene with high spatial resolution. The ideal MS data are obtained by SRF, which is from GF1. The subregions of 610 × 100 × 103 are picked for training (red regions), and the remaining regions are used for testing, as shown in Fig. 7(f).

B. Implementation Details and Evaluation Metrics

The following settings are used in all experiments. In the training phase, 32 × 32 overlapping HS–MS sample pairs are randomly selected and put into the network following data augmentation. We set the number of SFEMs to one and the number of DFEMs to two. In USTM, the depth of the U-shaped module is set to three, and the number of transformer blocks at each level is set to one. The proposed CTJN is executed on the Pytorch framework. The batch size is set to 16, and the optimizer is chosen Adam. The learning rate is initialized as 0.0001, and the decay policy is chosen cosine annealing scheme. It takes approximately 40 h to stabilize after 300 epochs for training on a Tian Xp GPU. In the test phase, we test the performance of different models on the whole image. Specifically, we set the step size to 16 and crop the testing regions into 32 × 32 overlapping MS patches. After reconstructing the MS patches into the corresponding HS patches, the edges of length eight are removed around the patches, followed by image stitching to achieve better results.

In the similarity-based evaluation, we apply six quantitative metrics to objectively evaluate the performance of different methods. The metrics are mean relative absolute error (MRAE), root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), structural similarity index (SSIM), and erreur relative globale adimensionnelle de synthèse (ERGAS). Besides, we sketch the spectral curves of different methods in various regions as samples. In addition, we visualize the performance of different methods qualitatively by error maps.

In the classification-based evaluation, we use four quantitative metrics to evaluate the classification results of SVM. The metrics are per-class accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). Besides, we use the McNemar test as a statistical analysis method to check whether the proposed CTJN enhances the classification performance statistically [67], which could be expressed as

$$\chi^2 = \frac{(f_{ij} - f_{ji})^2}{f_{ij} + f_{ji}} \tag{17}$$

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Fig. 7. Six HS–MS datasets for experiments. (a) HS200501. (b) HS200628. (c) HS200724. (d) HS200926. (e) HS210605. (f) PU.

TABLE III
PERFORMANCE OF OUR MODEL WITH DIFFERENT TRANSFORMER STRUCTURES ON EXPERIMENT 3'S DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Encoder	Decoder	MRAE	RMSE	PSNR	SAM	SSIM	ERGAS
RTB	RTB	0.0331	0.0132	43.5723	2.2894	0.9901	4.7520
HF-TB	RTB	**0.0300**	**0.0117**	**44.5789**	**1.9796**	**0.9916**	**4.3211**
RTB	HF-TB	0.0340	0.0135	43.3167	2.3386	0.9896	4.8823
HF-TB	HF-TB	0.0317	0.0127	44.2587	2.1057	0.9898	4.4757

where f_{ij} is the number of samples predicted correctly by the ith method but wrongly by the jth method. At the 95% confidence level, it can be considered that these two methods have significant differences when $|\chi| > 1.96$.

C. Ablation Analysis

In this section, we perform comprehensive ablation experiments on the dataset from experiment 3 to analyze the reconstruction capability of different CTJN parts.

1) Analysis of HF-TB: In USTM, we design four structures to evaluate the performance of HF-TB. The specific details and the performance are shown in Table III. It can be shown that utilizing HF-TB in the encoder stage and RTB in the decoder stage provides the greatest SR capability, followed by using HF-TB in both the encoder and decoder stages, and producing the poorest result is using RTB in the encoder stage and HF-TB in the decoder stage. Using HF-TB in the encoder stage could better maintain the image's fine details, resulting in reconstructed images that are closer to the actual HS images, while the decoder does not require additional high-frequency information to highlight.

2) Analysis of the Location of the Self-Attention Block Embedding: To evaluate the impact of the self-attention mechanism of the model, we design three locations for the self-attention block embedding, including using DFEM alone (embedding self-attention blocks in the entire network), using DFEM in the first half and SFEM in the second half of the network (embedding self-attention blocks in the first half of the network), and the proposed CTJN (embedding self-attention blocks in the second half of the network). The results are shown in Table IV.

It can be seen that the best results are achieved by CTJN, which uses SFEMs in the first half of the network to extract both spatial and spectral local features and DFEMs in the second half to extract the global features of the spectrum. When the self-attention mechanism is used in the first half of the network or the entire network structure, the model will directly capture the intercorrelations between long-range bands, thus allowing the model to ignore detailed information, making the reconstructed images too smooth and differing significantly from the actual images in regions where features change drastically.

TABLE IV
ABLATION ANALYSIS OF THE LOCATION OF THE SELF-ATTENTION MECHANISMS ON EXPERIMENT 3'S DATASET. THE BEST RESULTS ARE MARKED IN BOLD

	MRAE	RMSE	PSNR	SAM	SSIM	ERGAS
DFEM	0.0336	0.0136	43.2083	2.3404	0.9891	4.9267
DFEM+SFEM	0.0319	0.0129	44.2659	2.1972	0.9900	4.6538
CTJN	**0.0300**	**0.0117**	**44.5789**	**1.9796**	**0.9916**	**4.3211**

TABLE V
PERFORMANCE OF OUR MODEL WITH DIFFERENT ATTENTION BLOCKS ON EXPERIMENT 3'S DATASET. THE BEST RESULTS ARE MARKED IN BOLD

	MRAE	RMSE	PSNR	SAM	SSIM	ERGAS
Baseline	0.0319	0.0128	44.2539	2.1235	0.9901	4.5048
SE	0.0315	0.0128	44.2675	2.0983	0.9903	4.4203
ECA	0.0317	0.0131	44.1219	2.1135	0.9903	4.4615
CBAM	0.0308	0.0122	44.3128	2.0547	0.9907	4.3811
SSRB	**0.0300**	**0.0117**	**44.5789**	**1.9796**	**0.9916**	**4.3211**

3) Analysis of SSRB: In order to verify the necessity of SSRB, we compare it with similar structures such as SE, ECA, and CBAM. The results are shown in Table V. It is clear that SSRB achieves the best results in all evaluation metrics, which show a 6.3% increase in MRAE and a 9.4% increase in RMSE compared to the baseline network, followed by CBAM, which also utilizes neighboring information in both spatial and spectral dimensions. The performance of ECA is comparable to SE and inferior to CBAM, demonstrating that it is necessary to employ the neighboring relationship in the spatial dimension. SSRB utilizes 1-D convolution and 3×3 convolution to explicitly aggregate the information of neighboring pixels and adjacent spectra via a dual-branch structure, which enables the model to perform better while having good interpretability.

D. Evaluation Based on Similarity

In four experiments, we compare CTJN with seven SOTA methods that are all based on deep learning, among which SSJSRN [28], physics-informed deep adversarial spectral synthesis (PDASS) [33], SSRAN [27], SR network based on

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TABLE VI
COMPARATIVE RESULTS OF DIFFERENT METHODS IN EXPERIMENT 1. THE BEST RESULTS ARE MARKED IN BOLD

	MRAE	RMSE	PSNR	SAM	SSIM	ERGAS
SSJSRN	0.0306	0.0116	42.1503	2.2647	0.9884	5.9920
PDASS	0.0297	0.0108	41.9171	2.0625	0.9833	5.7130
SSRAN	0.0253	0.0086	43.9221	1.7064	0.9916	4.4786
SRT	0.0271	0.0100	42.6969	1.8843	0.9903	5.1502
PSJN	0.0215	0.0077	44.9977	1.4686	0.9931	4.1865
AWAN	0.0223	0.0078	45.0664	1.4855	0.9933	4.2391
MST++	0.0226	0.0078	44.5278	1.4920	0.9928	4.2323
CTJN	**0.0208**	**0.0072**	**45.3941**	**1.3566**	**0.9941**	**3.8794**

transformer and ResNet (SRT) [29], and PSJN [19] are proposed for remote sensing, and AWAN [54] and MST++ [43] are proposed for computer vision.

The data utilized in the training set and test set of experiment 1 are identical in terms of acquisition time and collection place. The distribution of ground features in the training and testing regions is similar. Table VI lists the reconstruction results of different methods. It can be seen that the accuracy of the reconstruction results of SSJSRN and SSRAN, which are designed based on small MS datasets, has poor performance. These two networks are lightweight, and it is challenging for lightweight networks to learn the precise mapping relationship between MS images and HS images in large-scale scenes with complex ground features and severe spectral mixing. For the remaining methods, PDASS also exhibits limited performance, which performs SR using spectral unmixing and a spectral library. The feature classes lacking in the spectral library will inevitably emerge in large-scale scenes, leading to a degradation of the effect. SRT uses a transformer to obtain global information in the spatial dimension of image blocks. However, global spatial correlations are unnecessary in large-scale scenes with low spatial resolution. Thus, SRT exhibits suboptimal performance in experiment 1. AWAN and MST++ exhibit comparable performance. PSJN surpasses AWAN regarding MRAE, RMSE, SAM, and ERGAS but is inferior in PSNR and SSIM. Our model achieves optimal results for all six quantitative metrics. Compared with the second-best method, the performance of our model is improved by 3.3% in MRAE, 6.9% in RMSE, 8.3% in SAM, and 7.9% in ERGAS. The proposed CTJN can explore the local spatial and global spectral features while entirely using prior knowledge to enhance the quality of the reconstructed images.

To objectively verify the efficacy of the proposed method, we perform independent testing in terms of feature classes. The regions selected for each class are shown in the boxes in Fig. 7(b). Table VII shows the results of various methods in different feature classes. It can be seen that our proposed method has the best results in most feature classes with SAM metrics, with only a slight difference from the best results in the suaeda class. CTJN can obtain good results in regions with various feature classes, demonstrating its efficacy in all classes.

Fig. 9 shows the error maps of various approaches on three uniformly chosen bands (559.2, 731.0, and 902.7 nm). The contents of the regions in the yellow boxes are enlarged to show the differences more clearly. It can be seen that the overall error of our model is less, demonstrating that our model's results are more accurate.

TABLE VII
COMPARATIVE RESULTS OF DIFFERENT METHODS WITH SAM METRICS ON THE PARTIAL EXPERIMENT 1 DATASET. THE BEST RESULTS ARE MARKED IN BOLD

	SSJSRN	PDASS	SSRAN	SRT	PSJN	AWAN	MST++	CTJN
Water	1.7563	1.5706	1.5513	1.4383	1.3927	1.565	1.3878	**1.3191**
Reed	3.5758	3.3527	3.296	3.2316	3.5598	3.5771	3.0187	**2.8516**
Mudflat	4.1639	2.9792	1.5076	3.6929	0.8716	0.9831	1.7492	**0.8656**
Tamarix	3.0078	2.8376	2.8628	2.7405	2.4554	2.4929	1.9864	**1.9511**
Suaeda	3.0476	2.812	**2.1738**	2.5708	2.4625	2.2824	2.6184	2.3480
Spartina	6.6356	2.8387	3.5715	3.1013	1.7214	2.2710	2.3350	**1.5446**



(a) (b)

Fig. 8. Details of training sets and test sets. (a) Training set for classification based on MS200628. (b) Test set for classification based on MS200628.

The 10 000 points with the most significant advantage of our method are counted in the reconstructed HS image. The white pixels in Fig. 10 are the most advantageous regions. It is observed that these pixels do not concentrate on a single specific ground feature class, showing that all models learn the information of various ground feature classes. At the same time, it is intuitively seen that white pixels are more concentrated in regions where ground features change drastically, and these regions represent the high-frequency information of the HS image. Our model extracts both spectral and spatial local features in SCM and emphasizes the high-frequency information in USTM, which gives the reconstructed HS image rich details and excellent performance in regions with drastic feature changes.

Fig. 11 shows the differences between the reconstructed spectral and ground-truth curves for representative feature classes. Fig. 11(a) and (b) shows the spectral curves of water and bare soils, respectively. It can be seen that most of the methods have good performance for the regions with clear feature classes, and our curves have the highest correlation coefficients with the ground truth. Fig. 11(c) and (d) shows the regions with severe spectral mixing, and it can be seen that CTJN can still achieve a more accurate reconstruction in these regions.

The test set used in experiment 2 is the same as the training set regarding acquisition time and place. Ground features have a similar distribution in both the training and testing regions. It is easier to learn the MS–HS mapping relation compared to experiment 1 since there is no thin-cloud interference. We reduce the training set for experiment 2, and Table VIII

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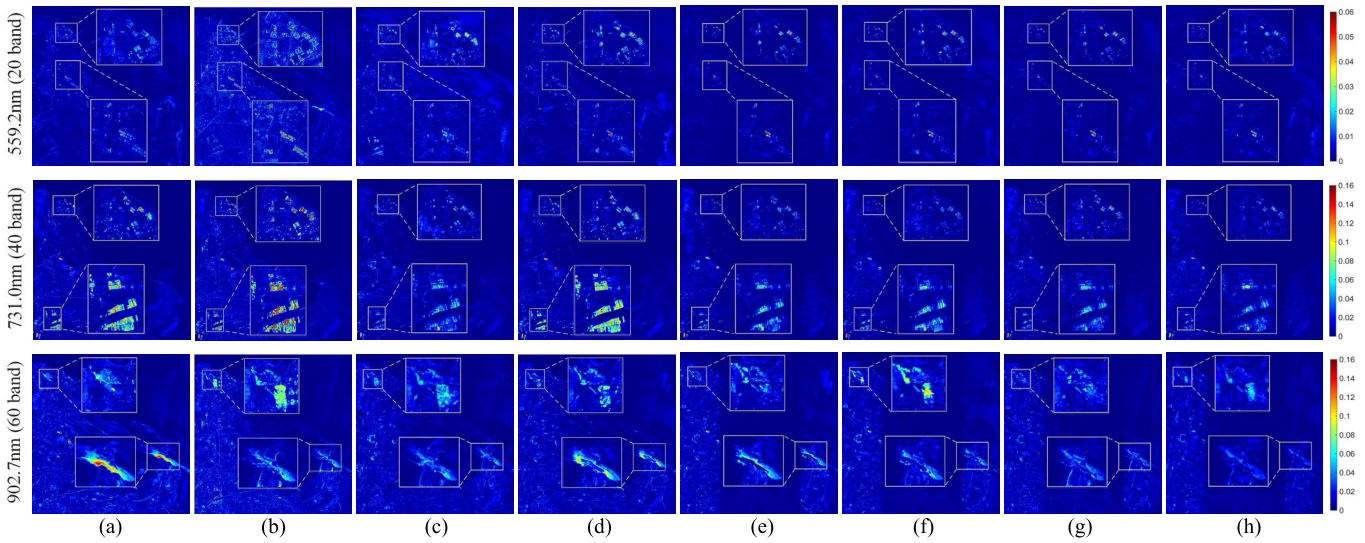


Fig. 9. Error maps of different methods on the HS200628 dataset in experiment 1. (a) SSJSRN. (b) PDASS. (c) SSRAN. (d) SRT. (e) PSJN. (f) AWAN. (g) MST++. (h) CTJN.

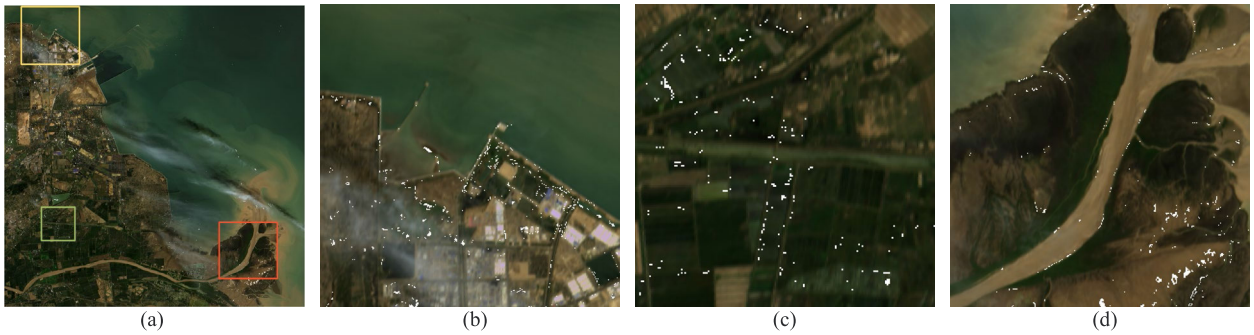


Fig. 10. 10 000 points (white pixels) of the greatest advantage of the SR of our model in experiment 1. (a) Statistics on the whole image of HS200628. (b) Enlargement of the ground features in the yellow box. (c) Enlargement of the ground features in the green box. (d) Enlargement of the ground features in the red box.

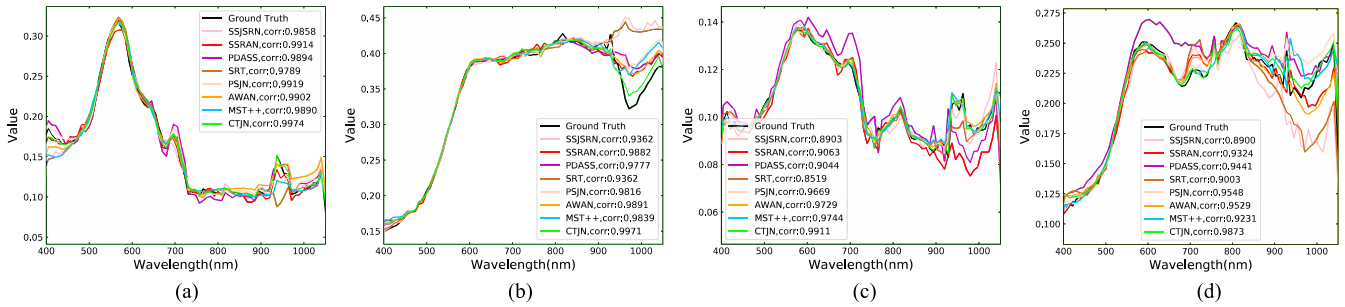


Fig. 11. Spectral curves and correlation coefficients of different methods for representative classes in experiment 1. (a) Pixels of water. (b) Pixels of bare soils. (c) Pixels of water and bare soil spectral mixing. (d) Pixels of water and plant spectral mixing.

shows the performance of several SOTA methods. SSJSRN and SSRAN still exhibit poor performance as lightweight networks. PDASS has limited performance due to the scenes still containing a large number of feature classes that are absent from the spectral library. SSRAN and SRT exhibit comparable performance. AWAN and MST++ show similar performance in several metrics. Our model achieves optimal results in all metrics except PSNR, where it achieves the second-best result. Compared to AWAN, the results of our model show a significant improvement of around 6.7% in MRAE, 5.4% in RMSE, and 8.5% in SAM, with a slight reduction in PSNR. In a comprehensive comparison, CTJN outperforms other approaches in experiment 2.

Fig. 12 shows the error maps obtained by different methods. Our reconstructed HS image is more accurate in places with complicated ground features than competitors. On the perceptual quality measure, our model produces the best-quality reconstruction. Furthermore, Fig. 13 shows the 10 000 points with the greatest advantage of CTJN. Compared to experiment 1, these points are more focused in regions where the ground features change drastically since there is no cloud-related noise interference.

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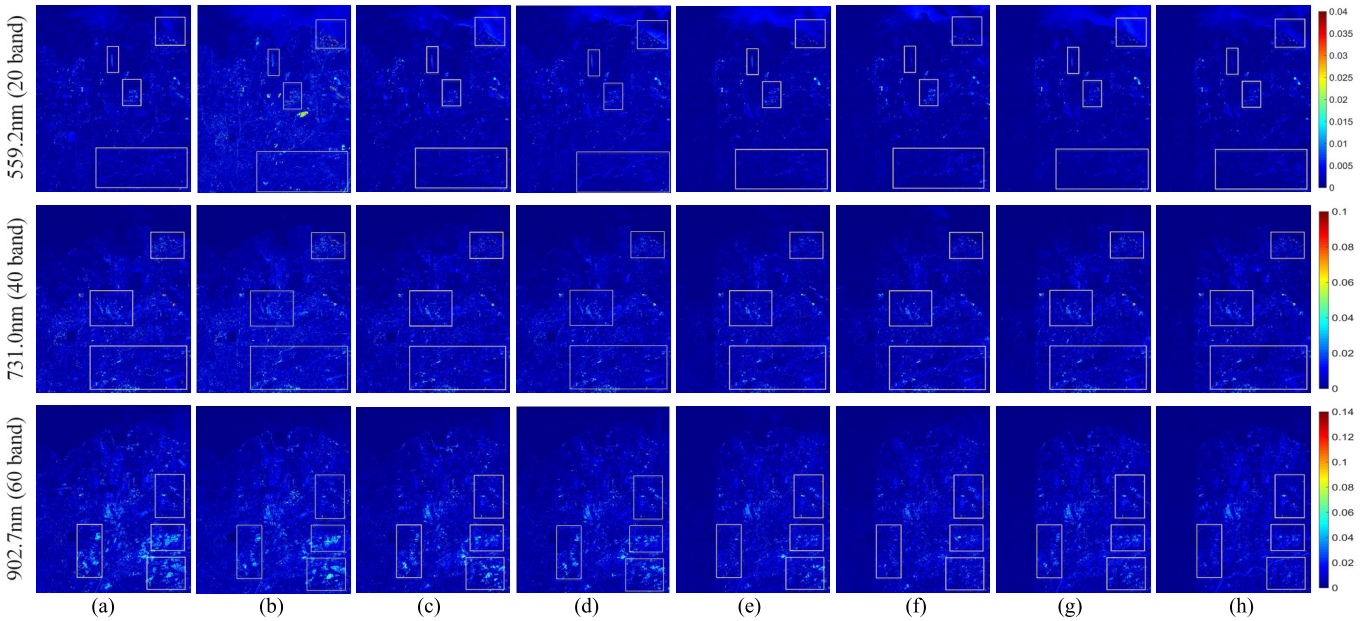


Fig. 12. Error maps of different methods on HS210605 dataset in experiment 2. (a) SSJSRN. (b) PDASS. (c) SSRAN. (d) SRT. (e) PSJN. (f) AWAN. (g) MST++. (h) CTJN.



Fig. 13. 10 000 points (white pixels) of the greatest advantage of the SR of our model in experiment 2. (a) Statistics on the whole image of HS210605. (b) Enlargement of the ground features in the yellow box. (c) Enlargement of the ground features in the red box. (d) Enlargement of the ground features in the green box.

TABLE VIII
COMPARATIVE RESULTS OF DIFFERENT METHODS IN EXPERIMENT 2. THE BEST RESULTS ARE MARKED IN BOLD

	MRAE	RMSE	PSNR	SAM	SSIM	ERGAS
SSJSRN	0.0223	0.0108	44.9953	1.6526	0.9905	3.7850
PDASS	0.0207	0.0104	45.1327	1.5476	0.9908	3.7571
SSRAN	0.0189	0.0094	45.2257	1.4182	0.9917	3.3269
SRT	0.0189	0.0095	46.1003	1.4243	0.9915	3.3222
PSJN	0.0168	0.0084	47.1040	1.2749	0.9927	3.0254
AWAN	0.0159	0.0077	**48.6523**	1.1742	0.9941	2.7974
MST++	0.0159	0.0079	47.7848	1.1655	0.9939	2.8697
CTJN	**0.0148**	**0.0073**	48.4260	**1.0813**	**0.9950**	**2.6829**

In experiment 3, the data utilized in the training and test sets are obtained at various times and in separate places. The results of different methods are listed in Table IX. Compared to experiments 1 and 2, it can be shown that the reconstruction results of all models have a certain degree of degradation.

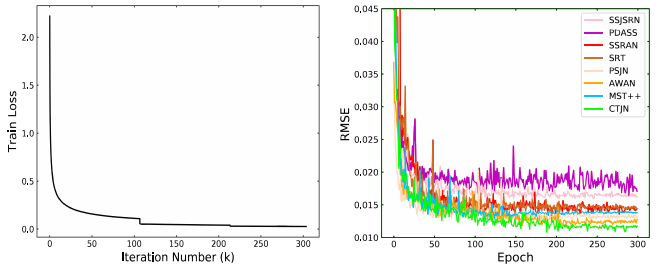


Fig. 14. Training process of CTJN on experiment 3's dataset. (a) Train loss of CTJN. (b) Convergence analysis of different methods.

The performance of CTJN still achieves the best performance in various quantitative metrics. Compared with the second-best method, CTJN performs a significant advance in MRAE, RMSE, SAM, and ERGAS metrics and a slight advance in PSNR and SSIM metrics.

Fig. 14(a) shows the learning progress of CTJN on the training set. The convergence of all methods on the experiment

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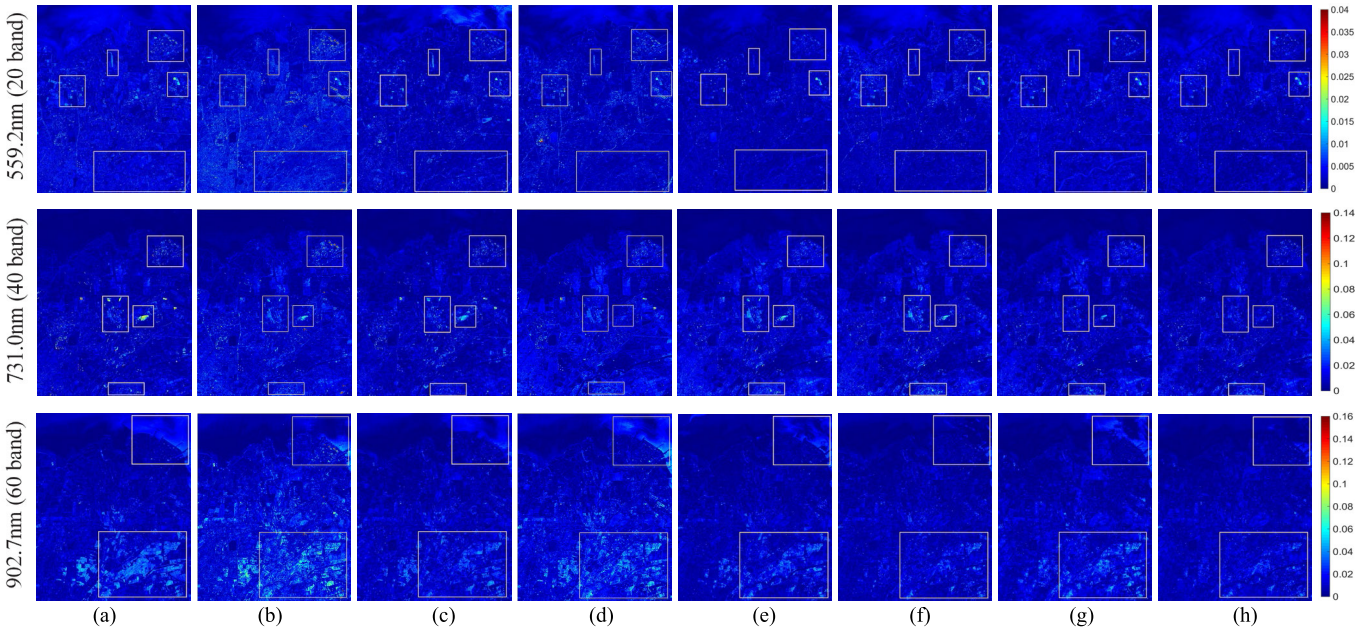


Fig. 15. Error maps of different methods on HS210605 dataset in experiment 3. (a) SSJSRN. (b) PDASS. (c) SSRAN. (d) SRT. (e) PSJN. (f) AWAN. (g) MST++. (h) CTJN.

TABLE IX
COMPARATIVE RESULTS OF DIFFERENT METHODS IN EXPERIMENT 3. THE BEST RESULTS ARE MARKED IN BOLD

	MRAE	RMSE	PSNR	SAM	SSIM	ERGAS
SSJSRN	0.0405	0.0164	41.4847	2.9868	0.9850	6.0344
PDASS	0.0387	0.0170	40.8886	2.6113	0.9863	6.0316
SSRAN	0.0346	0.0142	42.4382	2.4638	0.9885	5.1003
SRT	0.0363	0.0143	42.8485	2.5156	0.9868	5.2322
PSJN	0.0330	0.0130	44.1009	2.3126	0.9896	4.5476
AWAN	0.0312	0.0121	44.4103	2.0937	0.9902	4.3703
MST++	0.0337	0.0137	43.2455	2.3515	0.9901	4.9747
CTJN	**0.0300**	**0.0117**	**44.5789**	**1.9796**	**0.9916**	**4.3211**

TABLE X
COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE PU DATASET. THE BEST RESULTS ARE MARKED IN BOLD

	MRAE	RMSE	PSNR	SAM	SSIM	ERGAS
SSJSRN	0.1130	0.0122	39.2796	3.1160	0.9772	10.8324
PDASS	0.1292	0.0117	39.6479	2.9007	0.9753	8.7402
SSRAN	0.0933	0.0096	40.4230	2.5191	0.9811	7.3141
SRT	0.0926	0.0112	40.0010	2.5730	0.9805	8.2691
PSJN	0.0980	0.0096	42.0508	2.5484	0.9806	7.1135
AWAN	0.0926	0.0098	40.9514	2.5108	0.9814	7.4812
MST++	0.0983	0.0107	40.5112	2.6806	0.9800	7.8878
CTJN	**0.0917**	**0.0091**	**42.5184**	**2.4560**	**0.9900**	**6.9935**

3 dataset in RMSE is shown in Fig. 14(b), and it can be seen that all methods tend to stabilize as epochs increase. Fig. 15 represents several error maps of different methods, and it can be observed that, compared with experiment 2, the value of the error maps in experiment 3 is increased. It can be inferred that the accuracy of reconstruction results is significantly enhanced when the training set and the test set contain scenes with similar ground features. Overall, in experiment 3, our model maintains an advanced level of performance in various evaluation metrics, which shows that our model works well in scenes with complex ground features.

On the PU dataset, all methods could converge quickly. Table X shows the results of different methods. It can be seen that SSRAN shows comparable performance to AWAN on the small dataset. Insufficient data degrade the efficacy of AWAN and MST++, whereas lightweight networks, such as SSRAN and SSJSRN, are less affected. The proposed CTJN exhibits a leading performance in all metrics, including an enhancement of approximately 5.5% in RMSE, 2.2% in SAM, and 1.7% in ERGAS in the small and simple scene.

E. Evaluation Based on Classification

The classification accuracy can be used as an indirect metric to check the reconstruction effect when no corresponding HS image exists in [19], [26], and [28]. In this section, a variety of reconstructed models trained in different experiments are applied to MS200628 to evaluate the effectiveness of these models by classification.

All models obtained from experiments 1–3 are applied to the SR of the actual MS image. Tables XI–XIII list the quantitative measures on the actual MS image with different reconstruction methods in three experiments. It can be seen that in the comprehensive evaluation, most of the reconstructed HS images improve the classification performance, among which our model achieves the best result in all experiments. By calculating $|\chi|$ of the classification results of the SOTA methods with CTJN, it is shown that our model can improve the classification results in a statistical sense. In addition, our model also achieves optimal accuracy in several classes, which further validates its effectiveness.

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TABLE XI

CLASSIFICATION RESULTS OF DIFFERENT METHODS OBTAINED FROM EXPERIMENT 1 ON THE MS200628 DATASET. THE RIGHTMOST COLUMN IS THE CLASSIFICATION BASED ON THE ACTUAL MS IMAGE. THE BEST RESULTS ARE MARKED IN BOLD

	SSJSRN	PDASS	SSRAN	SRT	PSJN	AWAN	MST++	CTJN	MS
Water	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	100
Reed	91.82	91.20	92.44	90.89	92.13	88.82	90.16	**95.76**	96.48
Mudflat	76.52	77.74	78.63	78.18	78.85	**81.51**	77.52	80.40	72.65
Tamarix	54.09	57.14	57.14	57.36	59.54	58.56	59.11	**62.49**	52.45
Suaeda	52.93	50.90	54.74	53.39	55.08	**56.43**	53.84	54.51	51.58
Spartina	99.19	98.96	98.96	99.31	98.85	99.31	**99.42**	99.31	99.19
OA	78.72	78.97	79.98	79.49	80.41	80.39	79.64	**81.81**	78.40
AA	79.09	79.32	80.32	79.85	80.74	80.77	80.01	**82.08**	78.73
Kappa	75.35	75.87	77.75	77.63	77.24	77.48	76.68	**79.27**	74.37
$\lvert\chi\rvert$	11.06	10.60	8.55	9.61	6.99	5.46	9.23	-	11.37

TABLE XII

CLASSIFICATION RESULTS OF DIFFERENT METHODS OBTAINED FROM EXPERIMENT 2 ON THE MS200628 DATASET. THE RIGHTMOST COLUMN IS THE CLASSIFICATION BASED ON THE ACTUAL MS IMAGE. THE BEST RESULTS ARE MARKED IN BOLD

	SSJSRN	PDASS	SSRAN	SRT	PSJN	AWAN	MST++	CTJN	MS
Water	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	100
Reed	89.65	90.17	90.99	**94.62**	91.20	93.17	90.89	93.58	96.48
Mudflat	74.97	74.64	73.53	70.10	73.86	74.53	74.31	**76.97**	72.65
Tamarix	57.58	57.58	56.71	55.73	56.60	57.47	54.20	**60.09**	52.45
Suaeda	49.32	54.63	58.01	56.32	**62.87**	58.92	60.95	59.03	51.58
Spartina	98.26	97.34	98.38	98.38	97.58	**99.08**	98.38	98.85	99.19
OA	77.92	78.68	79.23	78.85	79.98	80.18	79.40	**81.10**	78.40
AA	78.30	79.06	79.61	79.19	80.35	80.53	79.79	**81.41**	78.73
Kappa	73.95	76.44	76.64	75.35	77.63	77.27	77.07	**78.55**	74.37
$\lvert\chi\rvert$	11.11	9.91	7.85	8.71	4.54	5.10	7.07	-	8.88

TABLE XIII

CLASSIFICATION RESULTS OF DIFFERENT METHODS OBTAINED FROM EXPERIMENT 3 ON THE MS200628 DATASET. THE RIGHTMOST COLUMN IS THE CLASSIFICATION BASED ON THE ACTUAL MS IMAGE. THE BEST RESULTS ARE MARKED IN BOLD

	SSJSRN	PDASS	SSRAN	SRT	PSJN	AWAN	MST++	CTJN	MS
Water	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	100
Reed	92.75	92.34	92.86	91.10	91.20	92.96	90.06	**94.51**	96.48
Mudflat	77.74	88.15	88.48	85.05	89.59	**92.03**	88.70	91.03	72.65
Tamarix	60.52	54.20	54.31	55.29	61.18	54.31	57.58	**63.47**	52.45
Suaeda	51.69	53.16	60.95	**64.22**	63.32	63.09	59.59	60.16	51.58
Spartina	98.85	98.27	98.96	98.86	98.15	98.85	**99.42**	99.31	99.19
OA	79.94	80.69	82.26	82.06	83.59	83.22	82.21	**84.49**	78.40
AA	80.26	81.02	82.59	82.42	83.91	83.54	82.56	**84.75**	78.73
Kappa	77.87	78.07	80.10	79.38	80.57	79.29	80.91	**82.10**	74.37
$\lvert\chi\rvert$	13.55	12.46	8.84	8.03	4.17	4.81	9.54	-	15.07

Comparing the classification evaluation of the three tables, it can be seen that the models in experiment 3 have the highest accuracy, followed by the models in experiment 1, and the models in experiment 2 have the worst accuracy. The results of McNemar's test between experiments 1 and 2 of CTJN could be shown as $|\chi_{21}| = 2.92$, and the results between experiments 1 and 3 could be shown as $|\chi_{13}| = 9.53$, which can be considered that the difference is significant. It implies that the classification accuracy is higher when the training set contains similar scenes to MS200628. The performance of the models still depends on the similarity between the training and test sets.

TABLE XIV

TIME COST OF DIFFERENT METHODS ON THE MS200628 DATASET. THE BEST RESULTS ARE MARKED IN BOLD

	SSJSRN	PDASS	SSRAN	SRT	PSJN	AWAN	MST++	CTJN
Time/s	134.74	33.14	**20.86**	95.11	788.40	73.00	92.98	167.69

F. Computational Cost

We test the time spent by different methods on the MS200628, which has approximately 2 million pixels. During the test phase, SRs are performed on a GTX 1650 GPU. The time costs for different methods are shown in Table XIV. The SSRAN takes the shortest at 20.86 s, while the PSJN consumes the longest at 788.40 s. For the same number of parameters, 3-D-CNN requires a great deal of processing time. SSJSRN and PSJN have 3-D-CNN cascade structures, substantially increasing the time needed for reconstruction. Our model uses a single-stage SCM to extract shallow features, which balances computational efficiency and model performance well. Considering that SR is generally regarded as an offline process in the current [19], [26], [48], [50], [68], [69], a portion of the sacrifice in computational cost is worthwhile in light of the improvement in reconstruction accuracy.

V. CONCLUSION

In this article, we propose a CTJN for SR in large-scale scenes with complex ground features from satellite MS imagery. The CTJN is composed of SFEMs followed by DFEMs. Specifically, the SFEM is composed of an SCM with a residual connection, in which the 3-D convolutions are utilized to extract local spatial and spectral features. The DFEM consists of a USTM with a residual connection, where the transformer blocks are applied to extract the global spectral features. Besides, we introduce HF-TBs for the transformer to preserve the detailed features of images, which could highlight the high-frequency information. Moreover, we design SSRB by prior knowledge to emphasize the correlation among neighboring pixels and adjacent spectra. Extensive experiments on four HS–MS datasets and one MS dataset show that CTJN has the most advanced performance compared to other methods, demonstrating the effectiveness of our model. In future work, we will focus on situations in which the image acquisition time is short, such as airborne sensors, and consider network reconfiguration time as a crucial criterion.

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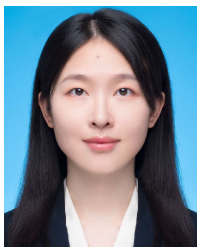
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